Geography Class 01

Geography(6:15 PM)

- Cosmology includes Galaxies, stars, Sun, and Earth.
- Physical Geography includes Geomorphology, Climatology, Oceanography, and Biogeography.
- Economic geography includes Resouces, Agriculture, Industries, and Transportation.
- Social or Human Geography includes the Population, Settlement, and Culture of people.
- An ecological aspect includes Ecosystems, Biomes, Types of vegetation, biodiversity, Pollution, etc.

Syllabus of geography(6:27 PM)

- ***Prelims:***
- Indian and World Geography - Physical, Social, Economic Geography of India and the World.
- ***Mains:***
- Salient features of the world's physical geography.
- Distribution of key natural resources across the world (including South Asia and the Indian sub-continent); factors responsible for the location of primary, secondary, and tertiary sector industries in various parts of the world (including India).
- Important Geophysical phenomena such as earthquakes, tsunamis, Volcanic activity, cyclones, etc., geographical features and their location changes in critical geographical features (including water-bodies and ice-caps) and flora and fauna and the effects of such changes.

PYQ sample discussion (6:47 PM)

Sources and reference for geography (7:09 PM)

- Have limited sources.
- ***Primary NCERTs:***
- Fundamental of Physical geography.
- India: Physical environment.
- Fundamental of Human geography.
- India people and economy.
- GC-Leong.
- Total geography of 10th class ICSE.
- Class notes(main source).
- VisionIAS value-added material.

Mapping (7:20 PM)

- ***Locations to be remembered:***
- Static location.
- Current affair-based location.
- Atlas.

Earth, solar system, and Universe (7:51 PM)

- Geography is made up of two words Geo+Graphy which means the description of the earth.
- The term geography was coined by Eratosthenes a greek philosopher called the father of geography.

- Geography is the study of places and relationships between people and their environment.
- ***The shape of the earth:***
- The earth is Flattened at the pole and bulged at the equator.
- It is due to the continuous rotation of the earth.
- This shape is called a geoid or oblate spheroid.
- Neuton proposed this for the 1st time and said the same applies to other planets too.

<u>The topic for the next class: The Shape of the Earth, longitude, and latitude.</u>

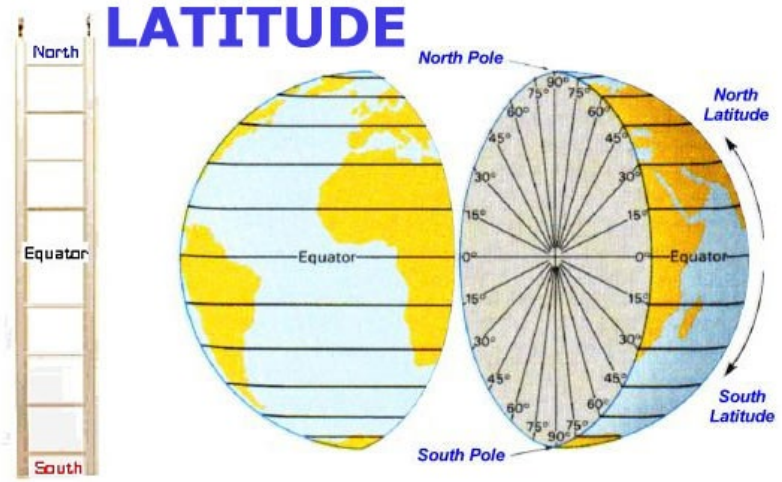
Geography Class 02

<u>The shape of the earth (5:10 PM)</u>

- The earth is Flattened at the pole and bulged at the equator.
- It is due to the continuous rotation of the earth.
- This shape is called a geoid or oblate spheroid.
- ***Evidence of the geoid shape of the earth:***
- Circumnavigation of the earth- Magellan completed the first circumnavigation in 1519.
- Circular horizon- The horizon appears circular when viewed from a high vantage point and the horizon widens with the increase in altitude.
- Ship's visibility- A ship appears to be rising from water when viewed from the coast or a ship(Bedford level experiment also proves the spherical shape).
- Sunrise and sunset- Different timing of sunrise and sunset at different locations.
- Eclipse- The earth's circular shadow falls on the moon during a lunar eclipse.
- Other planetary bodies- Since all other planetary bodies are spherical, the earth should also be spherical since the earth is also a part of the same solar system.

<u>Latitude and longitude (5:46 PM)</u>

- ***Latitude:***
- Diagramatic representation of latitude:



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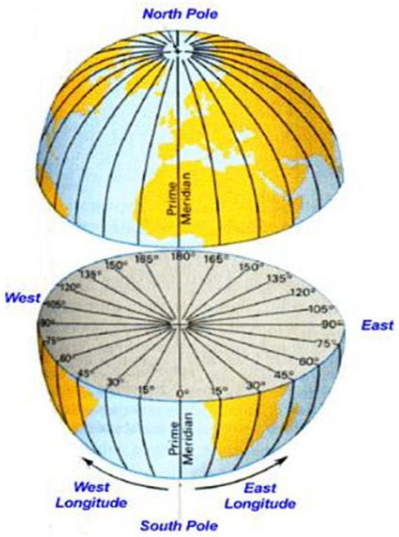
- The angular distance of the point on the earth's surface measured in degrees from the centre of the earth towards north & south of the equator is called the Latitude.
- Parallels of latitudes are the imaginary lines connecting places with the same latitude.
- The largest parallel of latitude is zero degrees and is called the equator.
- The parallels of latitudes are always parallel to the equator and one another.

- The length of the latitude decreases from the equator towards the north pole and south pole, however, the distance between them remains the same.
- The distance between one degree of latitude is equal to 111 km anywhere on the earth.
- ***Longitude:***
- Diagramatic representation of longitude:



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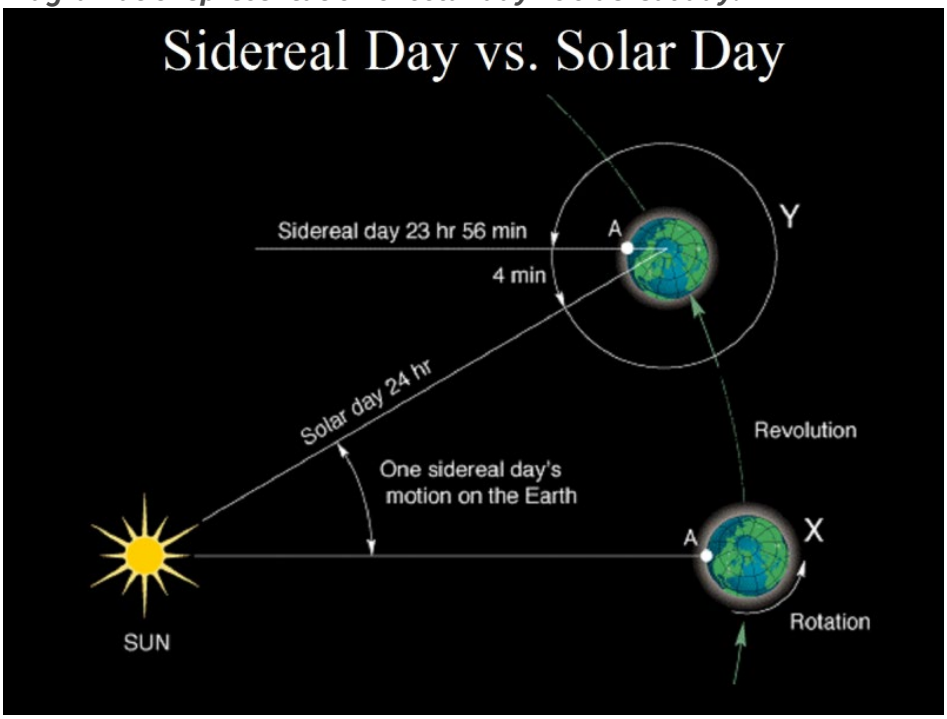
- Longitude is the angular distance of a place east or west of the prime meridian.
- Meridians are the semi circles running from pole to pole connecting places with the same longitude.
- The meridians of longitudes are not parallel to each other.
- The distance between them is maximum at the equator and decreases towards the pole.
- The distance between two meridians separated by 1 degree is equal to 111 km at the equator and it gradually decreases and is zero at the pole.
- The length of the meridian always remains the same.
- ***Great circle:***
- It is the longest possible circumference that can be drawn on earth.
- A great circle divides the earth into 2 equal halves.
- An infinite number of great circles can be drawn on the surface of the earth.
- A Great circle is used to find the shortest distance between two locations on the surface of the earth.

<u>The rotation of the earth (7:02 PM)</u>

- The spinning motion of the earth on its axis is called the rotation of the earth.
- The axis of rotation is the imaginary line passing through poles and the centre of the earth around which the earth rotates.
- The orbital plane is the plane in which the earth orbits around the sun.
- The angle between the axis of rotation and the orbital plane is 66.5 degrees.
- The angle of tilt of the axis of rotation from its normal position is equal to 23.5 degrees.
- The direction of rotation is counterclockwise or West to East.
- ***Period of rotation:***

- ***Solar Day:*** The time taken by the earth to rotate on its axis so that the sun appears in the same position in the sky is called Solar day.
- The solar day is equal to 24 hours.
- ***Sidereal day:*** The time is taken for the earth to rotate on its axis so that a distant star appears in the same position in the sky is called a Sidereal day.
- The sidereal day is equal to 23 hrs 56min.
- ***Diagramatic representation of solar day Vs sidereal day:***



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- ***Speed of earth's rotation****:*
- The linear speed of rotation of the earth is the maximum at the equator and reduces towards the poles.
- Therefore rockets are launched closer to the equator so that it provides an initial truth during the launching.

<u>Revolution of the earth (7:43 PM)</u>

- Revolution is the movement of the earth around the sun.
- Earth revolves around the sun in an elliptical orbit.
- The period of the revolution of the earth is 365 days and 6 hrs.
- The average speed of the revolution of the earth is nearly 1 lakh Km/hr.
- The direction of the revolution is in the anti-clockwise direction.
- Perihelion is the position of the earth nearest to the sun (On 3rd January).
- Aphelion is the position of the earth farthest from the sun (On July 4th).

<u>Seasons (7:56 PM)</u>

- Geographical there are 4 seasons- Summer, Winter, Autumn, and Spring.

<u>The topic for the next class: The occurrence of Seasons on earth.</u>

Geography Class 03

<u>Last class revision (5:12 PM)</u>

- Latitude and longitude.
- Rotation and revolution.

<u>Seasons (5:25 PM)</u>

- ***Tabular presentation for various cases of earth's movement:***

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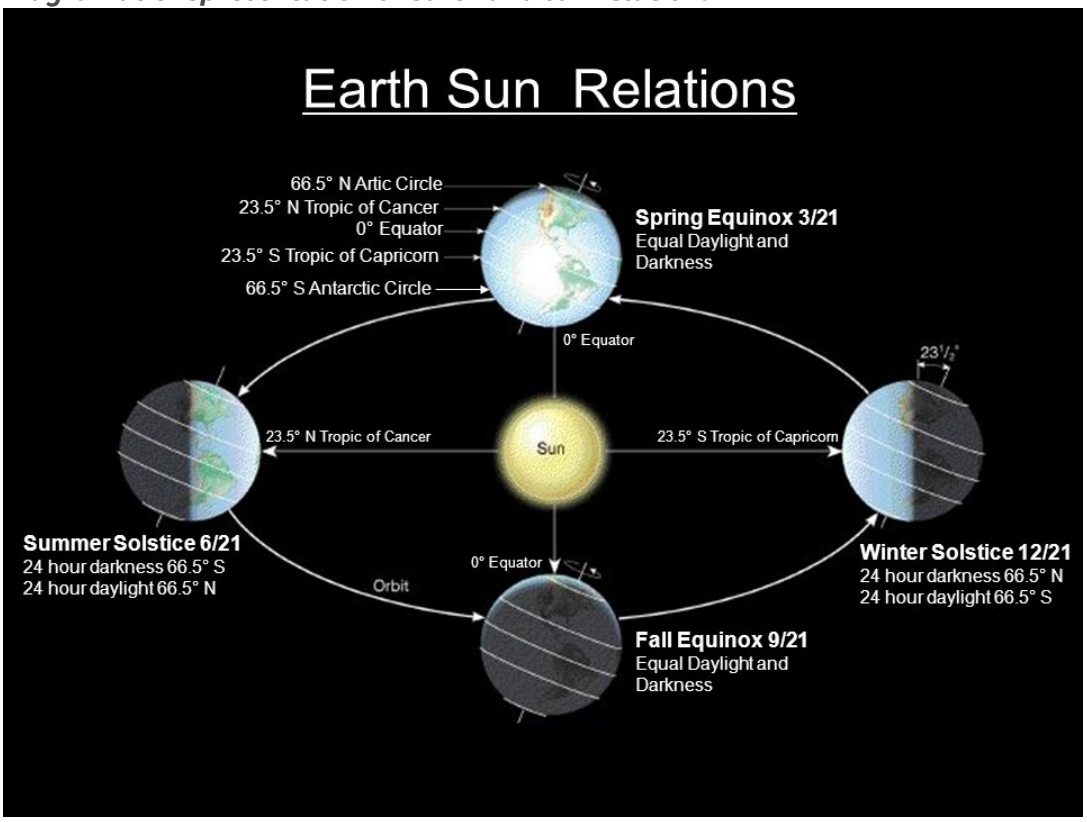
Conditions	The angle of incidence(intensity).	Length of the day
Case 1: No rotation. No tilt. No revolution.	90 degrees at the equator(higher intensity). 0 degrees at the poles(lower intensity).	24 hrs of day or night.
Case 2: Rotation is present. No tilt. No revolution.	90 degrees at the equator(higher intensity). 0 degrees at the poles(lower intensity).	12 hrs of the day and 12 hrs of the night across all the latitudes.
Case 3: Rotation is present. Tilt(northern hemisphere towards the sun) is present. No revolution.	90 degrees at 23.5 degrees north. The intensity is higher in the northern hemisphere than in case 2. In the southern hemisphere lower intensity than in case 2.	Equator- 12 hrs of day and night. The length of the day increases toward the northern hemisphere and decreases toward the southern hemisphere.
Case 4: Rotation, tilt, and revolution all are present.	90 Degrees at 23.5 degrees south. The southern hemisphere has a higher intensity than case 2. The northern hemisphere has a lower intensity than case 2.	Equator- 12 hrs of day and night. length of the day increases in the southern hemisphere and decreases in the northern hemisphere.

- Seasons are periods of the year that are characterized by specific climatic conditions.
- There are 4 seasons- Summer, Autumn, Winter, and Spring.
- Seasons are observed on the earth because the earth revolves around the sun with a tilted axis of rotation whose angle of inclination is always constant.

- This causes variations in the intensity of sunlight and length of day throughout the year resulting in seasons
- The regions receiving higher intensity of sunlight for a longer duration experience higher temperatures resulting in summer.
- Those regions receiving lower intensity of sunlight for shorter duration experience winters.
- ***Diagramatic representation of earth and sun relation:***



- ***Summer solstice:***
- It occurs on 21st June.
- The sun's rays fall vertically at the tropics of cancer(23.5 degrees N).
- The intensity of sunlight has increased in the northern hemisphere.
- The length of the day increases from the equator to the north pole.
- ***Winter solstice:***
- It occurs on 22 December.
- Sun rays fall vertically at the tropic of Capricorn(23.5 degrees S).
- The location of the southern hemisphere are receiving higher intensity of sunlight and those in the northern hemisphere lower intensity of sunlight.
- The length of the day decreases from the equator to the north pole.
- ***Equinox:***
- It happens on the 21st of March(Spring Equinox) and the 23rd of September(Autumn Equinox).
- The sun's rays are falling vertically at the equator.
- The length of the day is 12 hrs along all the latitudes.

Video related to seasons (6:31 PM).

The extent of day and night (6:49 PM)

- ***Position of the overhead sun:***
- It is a position where 90-degree sunlight falls.

- The position of the overhead sun is always between 23.5 Degrees N and 23.5 Degrees S.
- Every location between the Tropic of cancer and the Tropic of Capricorn will receive, at least 2 days of 90-degree sunlight and those which are located exactly in the tropics receive 90-degree sunlight only once.
- ***Polar day:***
- It is experienced in regions with 24 hours of daylight.
- The maximum extent of a polar day is 66.5 degrees N and S.
- ***Polar night:***
- It is when there are 24 hours of darkness.

Time Zones (7:20 PM)

- A time zone is a region of the globe that observes uniform standard time.
- The entire globe is divided into 24 time zones with the Greenwich meridian as a standard reference.
- The time along a particular longitude always remains the same.
- From the Greenwich meridian towards the east the time increases(EGA- east gain addition), and towards the west time decreases(WLS-west lose subtraction).
- With a change of every 15 degrees time changes by 60 mins, therefore with a change of every 1 degree, time changes by 4 mins.
- India follows 82.5 degrees east as Indian standard time.
- ***Multipule time zones:***
- India followed 3 time zones based on Bombay, Calcutta, and Madras, along with local Chaibagan time in Assam.
- In 1906 India adopted 82.5 degrees east as Indian standard time(IST) and the local time zones of Bombay and Calcutta were continued till 1955.
- The longitudinal extent of nearly 30 degrees between East and West has resulted in a Mismatch of the Sun cycle and human activities.
- ***Issues in adopting multiple time zones in India:***
- Economic integration- banking, stock exchange, etc.
- Administrative convince- offices, schools, etc.
- Issues in synchronization of transportation- Railways.
- Communication gap impacting security.

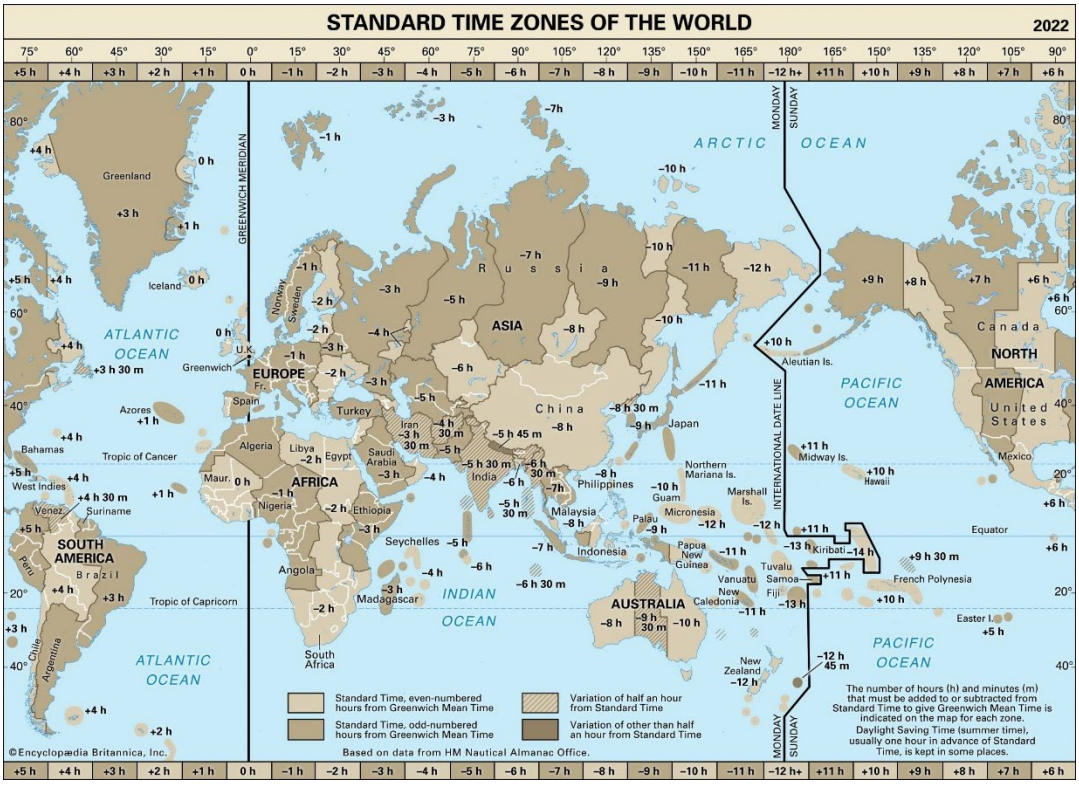
The topic of the next class: International Date Line and Origin of the Universe.

Geography Class 04

Last class revision (5:13 PM)

International date line (5:36 PM)

- ***Diagramatic representation of international date line:***



- It is an imaginary line of demarcation on the earth running from the North to South poles and demarcating the change of calendar day.
- It was agreed in 1884.
- IDL is not a straight line and follows a zig-zag path.
- If a person crosses IDL from West to East, she/he gains a day.
- If a person crosses IDL from East to West, she/he loses a day.

Daylight saving time (5:54 PM)

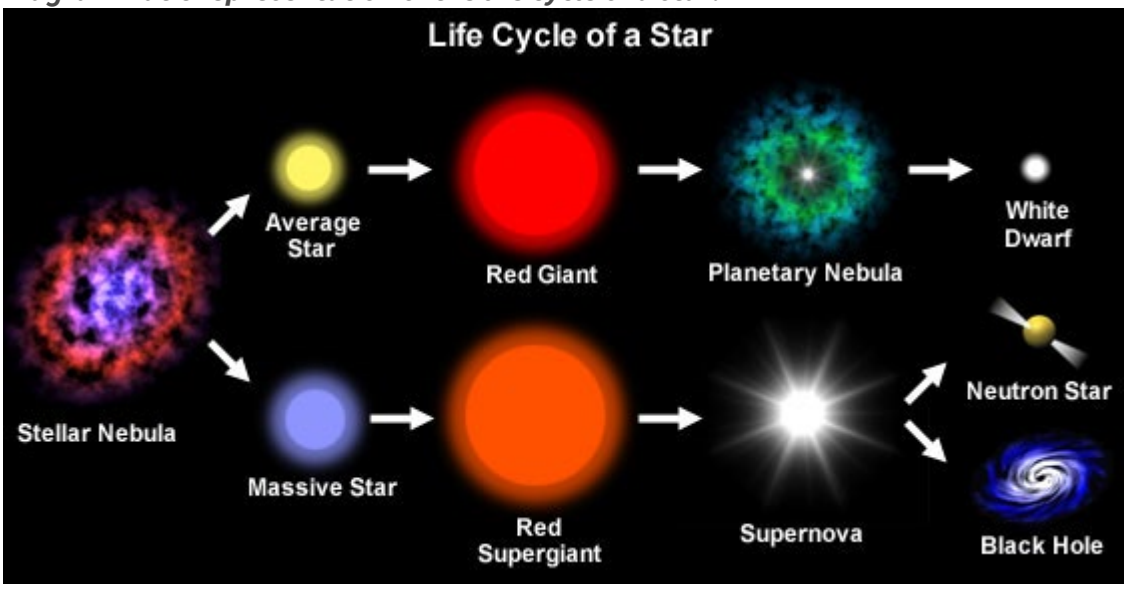
- It is the practice of shifting time by 1 hour ahead by forwarding the time in the clock just before the beginning of summer.
- It is practised in those regions where the length of the day varies between summers and winters.

Universe (6:05 PM)

- The limitless expanse of space surrounding us consists of the solar systems, stars, galaxies, etc.
- The Universe has around 100-400 billion galaxies with each galaxy having 100-400 billion stars.
- Origin of the Universe:
- ***Steady-state Theory:***
- It was proposed by Fred Hoyle.
- The overall size and mass of the universe remain constant at any point in time.
- The universe has no beginning and no end and is always expanding, creating new stars and galaxies at the rate old ones become unobservable.
- ***Pulsating Theory:***
- Proposed by Arthur Eddington.
- As per the theory Universe expands and contracts alternatively.
- ***Big Bang Theory:***
- It was proposed by Gorgeous Lemaitry in 1927.

- According to the theory, 13.7 billion years ago, the Universe was an extremely compact, dense, and hot singularity.
- There was no matter, no space, and no time.
- 13.7 billion years ago, a cosmic explosion happened called as Bing bang.
- From that time the universe started to expand and is continuing today.
- The expansion subsequently created the forces of Physics including Gravity, the formation of elementary particles, atoms, molecules, gaseous compounds, stars, and galaxies.
- Hubble, in 1929, proposed that all observable stars and galaxies are moving away from earth.
- He discovered this through the phenomenon of the Red Shift.
- The rate of expansion of the universe is called Hubble's constant.

<u>The life cycle of a star (7:16 PM)</u>

- ***Diagrammatic representation of the life cycle of a star:***
- 

- A nebula is a huge cloud of gas and dust mainly consisting of hydrogen.
- The gas is pulled together by gravity and the entire nebula starts to spin.
- The spinning gaseous mass reaches a high temperature resulting in the formation of a hot core called ProtoStar.
- When the temperature of the core crosses 15 million degrees, a nuclear fusion reaction begins at the core resulting in the birth of a star.
- When the supply of hydrogen runs out the core contracts and simultaneously the outer shell expands due to the fusion reaction resulting in Red Giant.
- If it is a low-mass star with less than 10 times the mass of the sun the core collapses leading to the formation of a planetary nebula
- If it is an average star with less than 10 times the mass of the Sun, the core of the Red giant collapses and leads to the formation of a Planetary nebula which is a spherical shell of gas.
- After some time, only the core is left inside the shell, which is called a White dwarf.
- A white dwarf becomes a Black dwarf over some time.
- If the mass of a star is more than 10 times the Sun, the star results in Red Supergiant.
- A Red Supergiant results in a supernova explosion due to the explosion of the iron core

- If the remnant core after the Supernova explosion is between 1.4 to 3 times the mass of the sun, it leads to the formation of a neutron star, which is a very heavy dense body consisting of closely packed neutrons.
- If the remnant core after the supernova explosion is more than 3 times the mass of the sun, the core collapses under gravity leading to the formation of a black hole.
- A black hole is a body with infinite gravity and density that even light can not escape from it.

The topic of the next class: Universe to continue.

Geography Class 05

Last class revision (5:17 PM)

Star and Galaxies (5:31 PM)

- ***Galaxy:***
- It is a sprawling system of gas, dust, and stars held together by gravity.
- All galaxies contain a supermassive black hole at the centre of the galaxy.
- **Types of galaxies:**
- ***Spiral galaxy:*** It is a spiral galaxy with spiral arms.
- It has a relatively flat disc appearance with a central bulge.
- ***Elliptical galaxy:*** It is a spherical or oval-shaped galaxy with a fairly uniform distribution of stars.
- ***Irregular galaxy:*** It has no definite shape or structure.
- ***Milkyway galaxy:*** It is a spiral galaxy with Sagittarius A black hole at its centre.
- Sun is located in Orion's arm of the Milkyway galaxy.
- Proxima Centauri is the nearest star to the sun.
- Sirius is the brightest star in the sky.
- Andromeda is the nearest galaxy to Milkyway.
- ***Twinkling of Stars:*** Due to the turbulence in the atmosphere, the light coming from stars is more deflected as it comes from a point source, However, planets do not twinkle.
- ***Light year:*** It is the distance travelled by light in one year at the speed of 3X10^8 m/s.
- ***Constellation:*** It is a group of stars forming a recognizable pattern.
- For Example The Big Dipper (Saptarshi Mandal)
- ***Pole star:*** It is the star that is aligned with the axis of rotation of the earth because of which it appears stationary throughout the night.
- In the northern hemisphere, it is Polaris, and south hemisphere it is Sigma Octantis.
- Pole start is always fixed in the sky however angle at which the pole star is visible varies with latitude.

Origin of the Solar system (6:23 PM)

- There are 2 types of theories to explain the origin of the solar system, Evolutionary and catastrophic theories.
- ***Evolutionary Theories:*** According to evolutionary theories the material of the solar system condensed into the sun and other planets simultaneously as isolated masses of matter from a single cloud of gas.
- Therefore both the Sun and the planets are of the same age.
- ***Gaseous hypothesis:*** It was proposed by Emmauinal Kant.
- There were cold and hard particles supernaturally created forming a gaseous cloud.

- The particles collide with each other due to gravitational attraction releasing heat.
- This gradually resulted in the formation of a rotating structure.
- Rings of matter were thrown off from this structure which cooled down to form planets.
- ***Nebular Hypothesis:*** It was proposed by Laplace.
- It is the most widely excepted theory.
- There was a pre-existing nebula in a rotating state.
- With gradual cooling, the nebula shrank which led to more spinning and resulted in the formation of a flat disk.
- The rings of matter got separated from this disk due to centrifugal force.
- The matter within the rings condensed to form planets.
- ***Catastrophic Theories:*** These theories are also called Binary theories since they assume the existence of a 2-star system.
- ***Planetesimal Hypothesis:*** It was proposed by Chamberlin and Moulton.
- The proto-star is accompanied by a companion star.
- This companion star came near the protostar the gravitational pull led to the ejection of matter called planetesimals.
- These planetesimals merged to form planets.
- ***Tidal Hypothesis:*** It was proposed by Genes and Jeffry.
- A very big intruding star came near to the sun and the gravitational pull led to the ejection of tides of material that condensed to form planets.

The topic of the next class: Solar system and earth.

Geography Class 06

Last class revision(5.15 PM).

Solar system(5.23 PM):

- ***Sun:***
- It is a dwarf star.
- The age of the sun is 4.8 billion years.
- Layers of the Sun:
- ***Core:***
- The innermost layer of the Sun's interior where nuclear fusion reaction happens.
- It has the highest temperature which is 15 million degrees C.
- ***Radiative Zone:***
- The energy from the Core is transferred outward by radiation through this zone.
- ***Convective Zone:***
- The energy is transferred through Convection currents to the surface.
- ***Photosphere:***
- It is the 1st layer of the Sun's atmosphere.
- Visible light originates from this zone.
- It is the brightest layer of the Sun.
- Its temperature is 5500 degrees C.
- ***Chromosphere:***
- It is the intermediate zone of the sun's atmosphere and emits colorful light.
- ***Corona:***
- It is the uppermost layer and is visible only during a total solar eclipse.
- It is hotter than the photosphere with a temperature of 2 million degrees Celsius.
- ***Solar Flare:***

- It is the sudden burst of energy with a storm of hot atoms released into space.
- ***Sunspots:***
- These are the dark spots in the Photosphere.
- They experience reduced temperature but strong magnetic activity.
- The Number of sunspots increases or decreases over some time.
- Solar maxima are when the sunspots are highest and solar minima are when they are lowest.
- The period between solar maxima or minima is 1 sunspot cycle(11 years).

Planets of our solar system(6.06 PM):

- They are of two types, Terrestrial planets (Earth-like) and Jovian Planets (Jupiter-like).
- ***Differences between the Terrestrial and Jovian Planets:***
-

Terrestrial Planets	Jovian Planets
Earth-like	Jupiter-like
High density	Low density
Rocky	Gaseous
Smaller in size	Large in size
High solar winds	Lower solar winds
High temperature	Low temperature
Thin/no atmosphere	Thick atmosphere
Less no. of satellites	Large no. of satellites
No rings	Rings
Speed of rotation lower	Very high speed of rotation

- ***Mercury:***
- It is the smallest planet.
- It is nearest to the sun.
- It has no atmosphere.
- ***Venus:***
- It is called Earth's twin.
- It is the hottest planet (very high greenhouse effect due to the presence of Co2)
- ***Earth:***
- It is called a Blue planet.
- It is the densest planet of all.

- ***Mars:***
- It is called a Red planet.
- the possibility of life due to its presence in Goldilocks Zones**(**habitable regions in the universe).
- ***Jupiter:***
- The largest planet in the solar system (11 times bigger than the earth).
- The giant red spot (a spinning cyclonic storm that appeared on Jupiter 400 years ago) is the most characteristic thing about Jupiter.
- ***Saturn:***
- It has the most prominent and spectacular ring system.
- It is the lightest planet in the solar system.
- ***Uranus:***
- Its axis of inclination is tilted by 98 degrees.
- It rotates from East to West(Retrograde rotation).
- ***Neptune:***
- It is the farthest planet.
- It cannot be seen with the naked eye.

Satellites(7.06 PM):

- It is a body that revolves or orbits around another planet (and not around the sun).
- Mercury and Venus have no satellites.
- Earth has one satellite - Moon.
- Mars has two satellites- Phobos and Deimos.
- Jupiter has 79 satellites, four of which are important (discovered by Galileo/ Galilean satellites) Io, Europa, Ganymede(the largest satellite), and Callisto.
- Saturn has the maximum number of satellites i.e. 82.
- One of the important satellites is Titan, which is the second-largest satellite in the solar system.
- Titan is one of the most probable regions in the solar system where there is the possibility of life's existence.
- Satellites of the Uranus are named after the characters of the play written by Shakespeare, for example, Miranda, and Oberon.
- Triton, a satellite of Neptune, rotates in the opposite direction of Neptune.
- ***Moon:***
- The period of rotation of the moon is equal to the period of revolution of the moon around the earth, which is equal to 27.3 days.
- Therefore the other half of the moon is not visible from the earth.

Dwarf planet(7.36 PM):

- In 2006, the International Astronomical Union set the conditions for a body to be declared a Planet:
- It should have an independent orbit around the Sun.
- It should have enough mass to achieve hydrostatic equilibrium i.e. spherical shape.
- It should have a cleared neighborhood.
- Officially there are 5 dwarf planets, there are the bodies that fulfill 1st 2 conditions but not the 3rd one, for example, Pluto, Eris, Ceres, Haumea, and Makemake.

Other important bodies in solar system(7.51 PM):

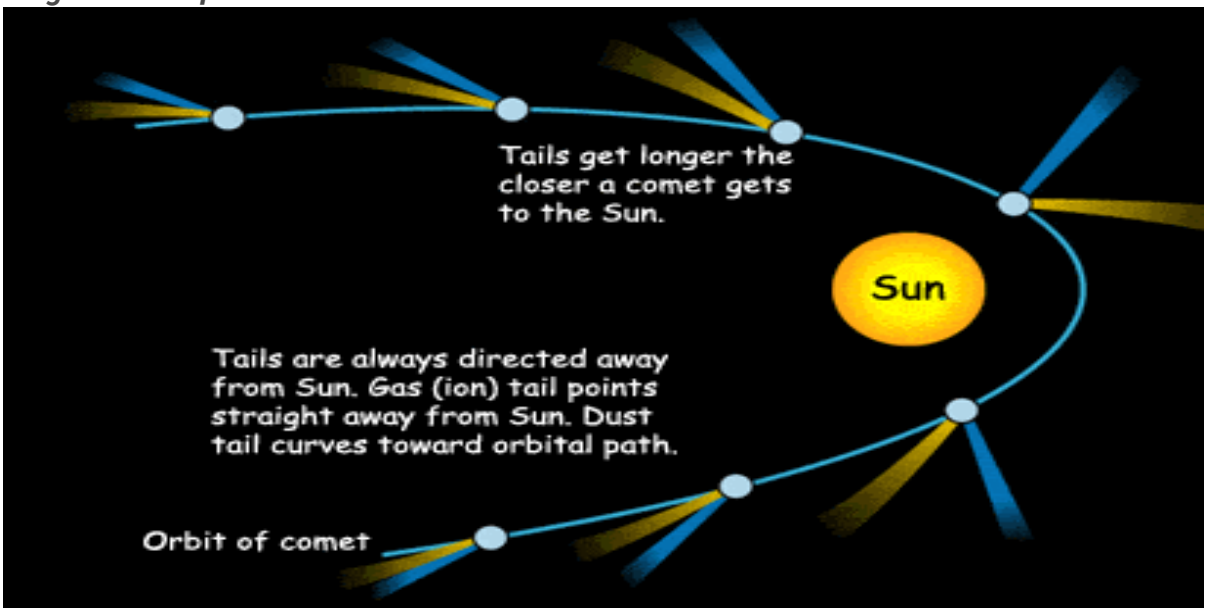
- ***Asteroid belt:***

- It is a belt of Asteroids which are small rocky and metallic bodies with cores made up of Iron and Nickel.
- They are orbiting around the Sun in a belt between Mars and Jupiter.
- ***Kuiper Belt:***
- This is a belt of remnants of planets with frozen gases, rocks, and debris materials.
- ***Oort cloud:***
- It is a layer of Ice pieces of space debris surrounding the entire solar system.
- ***Meteoroids:***
- When an asteroid comes out of its orbit and moves toward the earth.
- ***Meteor:***
- It is a flash of light due to the burning of meteoroids in the earth's atmosphere.
- It is also called a Shooting star.
- ***Meteorite:***
- It is the remnant of a meteoroid that has survived its passage through the earth's atmosphere.

<u>The topic for the next class: Comet, Solar, and Lunar eclipses.</u>

Geography Class 07

<u>Comet (5:20 PM)</u>

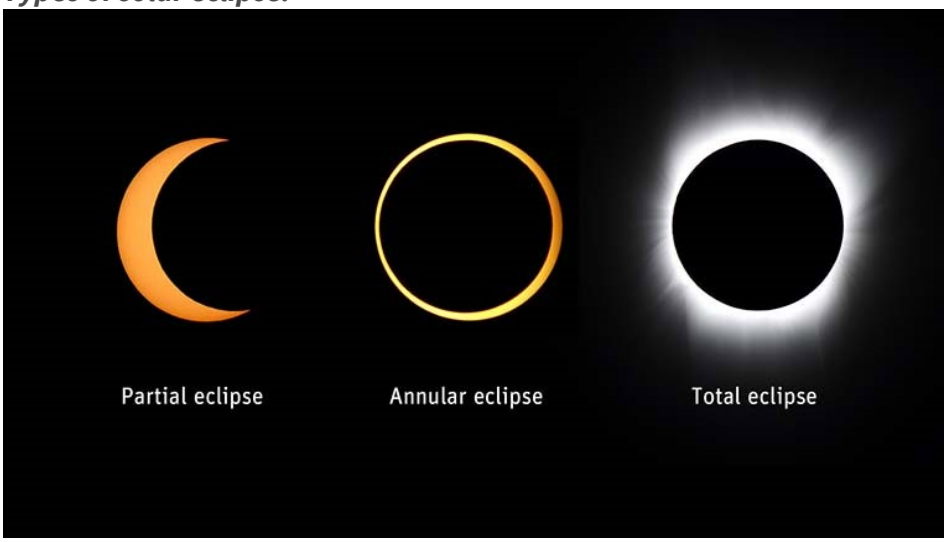
- ***Diagramatic representation of the orbit of Comet:***
- 

- It is a frozen, rocky core surrounded by gas and dust in a frozen state.
- They develop a very elongated orbit around the sun.
- The Comets originate from the Kuiper belt.
- As they come closer to the Sun, the Sun's heat melts the ages in Coma resulting in the formation of a tail pointing away from the sun.
- The tail is longest when the Comet is closest to the Sun.
- For example Halley's comet(1986), Comet Neowise(2022), Hale Bopp(1997), and Leonard(2021).

<u>PYQ discussion (5:37 PM)</u>

<u>Eclipse (5:39 PM)</u>

- When one heavenly body moves into the shadow of another it results in an eclipse.
- Umbra is the darker shadow of the blocking body
- Penumbra is a lighter shadow that causes a partial eclipse.

- ***Solar eclipse:***
- When the moon blocks the light of the Sun from reaching the earth, it casts a shadow onto the earth resulting in a solar eclipse.
- ***Types of solar eclipse:***
- 

- Total solar eclipse:
- It is when the sun is completely blocked by the moon.
- The photosphere is blocked and the Chromosphere and Corona will be visible.
- It is visible from the umbra.
- Partial solar eclipse:
- It is when the sun is partially blocked.
- It is visible from the penumbra.
- Annular solar eclipse:
- When the moon is farthest from the earth(apogee).
- It will not be able to block the entire photosphere creating a ring in the sky called the ring of fire.
- ***Lunar eclipse:***
- When the earth comes in between the sun and the moon blocking the light from the sun which was supposed to be reflected by the moon causing the earth's shadow to fall onto the moon.
- ***Type of lunar eclipse:***

-
- Total lunar eclipse:
- It is when the moon is located within the umbra of the earth and is totally shadowed.
- In this position, only the light from the earth's atmosphere reaches the moon which has only red light therefore the moon appears red.
- Partial lunar eclispe:
- When the moon is located between the Umbra and Penumbra, the earth's partial shadow covers the moon resulting in a partial lunar eclipse.
- Penumbral lunar eclispe:
- When the moon is completely located in the penumbra.
- The moon appears darker in comparison to the full moon.
- ***Various types of the moon:***
- Blood moon:
- Total lunar eclipse.
- Super moon:
- A full moon in perigee.
- Blue moon:
- It is the second full moon of the month.

The difference between the solar eclipse and the lunar eclipse (6:34 PM)

- The solar eclipse happens during a new moon while the lunar eclipse happens during a full moon.

- A solar eclipse can happen during the daytime and a lunar eclipse can happen only in the nighttime.
- The lunar eclipse is longer in duration as compared to the solar eclipse.
- You should not see the solar eclipse directly as it can damage the pupil of the eye due to sudden changes in sunlight a solar eclipse lasts for a very short duration while it is okay to see a lunar eclipse directly.

Mapping (7:17 PM)

- **Oceans:**
- 71% of the earth is covered with water.
- The oceans are interconnected with each other and don't have strict boundaries.
- ***There are 5 different oceans:***
- The Pacific ocean.
- The Atlantic ocean.
- The Indian ocean.
- The Southern ocean.
- The Arctic ocean.
- ***Pacific Ocean:***
- The Pacific Ocean is the largest and deepest of all the oceans.
- The pacific ocean is twice the size of the Atlantic ocean(2nd biggest ocean).
- The Pacific Ocean is connected to all other oceans.
- It is connected to the arctic ocean through the Bering Strait.
- The strait is a narrow water body connecting two big water bodies.
- The Drake passage also connects the pacific and Atlantic oceans.
- The Strait of Malacca connects the Pacific and Indian oceans.
- The pacific ocean is the deepest part of the earth that is the Challenger deep(10,900 meters) in the Mariana Trench near the Philippines.
- ***Atlantic Ocean:***
- It has an "S" shape.
- It is the youngest ocean of all oceans.
- It has a huge mid-oceanic ridge.
- Iceland is part of the mid-oceanic ridge.
- ***Indian Ocean:***
- It is the only ocean named after a country.
- No connection between the Indian and the Arctic oceans directly.
- ***Arctic Ocean:***
- It is the smallest ocean and remains frozen throughout the year.
- The impact of global warming is very high in the Artic ocean.
- ***Southern Ocean:***
- International Hydrographic Organization ratified the southern ocean as the 5th ocean.
- It is limited to 60 degrees south latitude.

Continents (7:53 PM)

- Listed in order of size the continents are:
- Asia> Africa>North America>South America,>Antarctica>Europe>Oceania.
- ***Asia:***
- The Ural Mountains are considered the border between Asia and Europe.
- The Caucasus mountains are the boundary of southern Europe and southern Asia.
- Asia is connected to Africa through the Sinai peninsula(part of Egypt).

- Caspian sea and the black sea as the water boundaries between Europe and Asia.
- Asia and Oceania have a land boundary in the new guinea islands.
- Mount Everest(located between Nepal and China) is the highest point on the earth in the Himalayas.
- Dead Sea(Between Israel and Jordan) is the lowest point on the land surface.

The topic for the next class: Geomorphology

Geography Class 08

Geomorphology (5:09 PM)

- It is the study of the physical features of the earth and the process by which those features are formed.
- ***Origin of the earth:*** Earth originated from a gaseous nebula.
- As the nebula cooled rings of matter ejected resulting in planets.
- ***Evolution of earth:*** The planet earth was initially barren, rocky, and in a volatile state.
- This primordial earth gradually evolved into the present stable one with a thick atmosphere.
- ***Formation of interior layers of the earth:***
- Due to a gradual cooling down of the earth, the heavier elements started to sink toward the centre and the lighter ones moved toward to surface through the process of density separation.
- With the gradual increase in density at the centre, the interior temperature increased
- With time as the earth cooled further, it condensed into a smaller size(condensation).
- The further process of differentiation led to the formation of different layers in the earth's interior.
- ***Evolution of atmosphere and hydrosphere:***
- The early thin atmosphere with hydrogen and helium was stripped off due to solar winds.
- During the cooling of the earth, gasses and water vapour were released from the earth's interior.
- And the process through which these gases outpour into the atmosphere is called Degassing.
- The volcanic eruption released more gasses and water vapour.
- The important gases at this stage were carbon dioxide, nitrogen, methane, water vapour, and every little oxygen.
- As the earth cooled further the water vapour started to condense and the CO_2 in the atmosphere got dissolved in rainwater and was brought down to the surface.
- This further reduced the temperature of the atmosphere which led to more condensation and more precipitation.
- The rainfall from the atmosphere got collected in the depressions on the surface forming oceans which were completed around 4000 million years ago.
- Life evolved in the oceans 3800 million years ago as Non-photosynthesis micro-organisms.
- Between 3000-2000 million years ago, blue-green algae emerged in ocean water which released oxygen through photosynthesis.